Differences in Posterior Ocular Morphology Following Red Light Therapy Versus 0.01% Atropine for Myopia Control: A Randomized Controlled Trial Analysis

Ziwei Zhao,[1,2] Yue Wu,[1,2] Fan Song,[1,2] Mingge Li,[3] Yueye Wang,[1,2] Ruoyu Chen,[1,2] Yanjun Liu,[1] Jinying Li,[4] Mingguang He,[1,2,5] and Yanxian Chen[1,2]

[1]School of Optometry, The Hong Kong Polytechnic University, Hong Kong SAR, People's Republic of China
[2]Research Centre for SHARP Vision, The Hong Kong Polytechnic University, Hong Kong SAR, People's Republic of China
[3]Department of Ophthalmology, Shenzhen Nanshan People's Hospital, Shenzhen, People's Republic of China
[4]Department of Ophthalmology, Peking University Shenzhen Hospital, Shenzhen, People's Republic of China
[5]Centre for Eye and Vision Research (CEVR), Hong Kong SAR, People's Republic of China

Correspondence: Yanxian Chen, The Hong Kong Polytechnic University, Hong Kong SAR, People's Republic of China; yan-xian.chen@polyu.edu.hk.

Received: February 5, 2026
Accepted: June 19, 2026
Published: July 14, 2026

Citation: Zhao Z, Wu Y, Song F, et al. Differences in posterior ocular morphology following red light therapy versus 0.01% atropine for myopia control: A randomized controlled trial analysis. *Invest Ophthalmol Vis Sci.* 2026;67(8):35. https://doi.org/10.1167/iovs.67.8.35

Purpose. The purpose of this study was to compare longitudinal changes in three-dimensional (3D) macular morphology between repeated low-level red-light (RLRL) therapy and 0.01% low-dose atropine (LDA) treatment in myopic children.

Methods. This secondary analysis of a randomized controlled trial (RCT) included 50 myopic children (aged 7–15 years) randomized to receive either RLRL therapy (650 ± 10 nm) or 0.01% atropine eye drops. Spectral-domain optical coherence tomography (OCT) was performed at baseline and 1, 3, 6, and 12 months. Custom software was used to perform distortion correction and quantify macular curvature (MC; mm^{-1}), macular outward scleral height (MOSH; µm), and their respective asymmetry indices. Primary outcomes were 12-month changes in average regional macular curvature. Secondary outcomes included changes in MOSH, asymmetry indices, and retinal layer thicknesses. Linear mixed-effects models analyzed longitudinal changes.

Results. At 12 months, RLRL showed greater macular flattening than 0.01% atropine across all regions (mean differences, $\times 10^{-3}$ mm^{-1}): central (9.0, 95% confidence interval [CI] = 4.4–13.6, $P < 0.01$), temporal (9.7, 95% CI = 6.7–12.8, $P < 0.001$), inferior (8.4, 95% CI = 5.2–11.6, $P < 0.001$), superior (7.0, 95% CI = 4.0–9.9, $P < 0.001$), and nasal (3.7, 95% CI = 0.8–6.7, $P < 0.05$). The inter-group difference was most pronounced centrally and decreased with eccentricity. MOSH decreased superiorly and increased inferiorly in the RLRL group at 12 months, whereas the LDA group showed minimal change. Both nasal-temporal and superior-inferior macular curvature asymmetry indices increased significantly more in the RLRL group ($P < 0.001$ and $P < 0.05$, respectively). No inter-group differences in retinal layer thickness changes were observed.

Conclusions. Over 12 months, RLRL showed broader macular flattening and more region-specific MOSH changes compared with 0.01% atropine in this cohort. However, the underlying mechanisms and the independent contribution of myopic progression to these morphological differences require further study.

Keywords: myopia control, repeated low-level red-light (RLRL), low-dose atropine (LDA), macular curvature, macular outward scleral height (MOSH), optical coherence tomography (OCT), posterior pole morphology

Myopia is a growing public health concern, particularly in East Asia, where its prevalence has reached epidemic proportions.[1] As a leading cause of visual impairment and blindness globally, myopia imposes substantial personal and societal burdens, highlighting the urgent need for effective interventions and a deeper understanding of structural changes occurring during myopia progression.[2]

Myopia progression is closely related to changes in posterior ocular morphology. Optical coherence tomography (OCT)-assessed macular curvature correlates significantly with both myopia severity and axial length (AL).[3–5] Moreover, steeper macular curvature in children associates with a sparser deep capillary plexus, suggesting a relationship between posterior morphology and retinal microvasculature.[6] Beyond curvature, "macular outward scleral height" (MOSH) quantifies posterior pole deformation in the macular region.[7–10] MOSH characterizes posterior scleral morphology across diverse populations, regardless of posterior staphyloma presence. Additionally, as myopic growth often occurs non-uniformly, analyzing shape asymmetry provides information beyond single

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average curvature or height summaries.[11] Moreover, in highly myopic eyes, the globe undergoes significant remodeling, characterized by posterior staphyloma formation, scleral thinning, and non-uniform axial elongation.[12–14] These morphological changes are significantly associated with pathological conditions, including myopic maculopathy, retinal detachment, and glaucoma.[15] Therefore, effective myopia control should address both progression and morphological changes to reduce sight-threatening complications.

Repeated low-level red-light (RLRL) therapy and 0.01% low-dose atropine (LDA) represent 2 interventions for myopia control. Recent head-to-head trials reported stronger slowing of both spherical equivalent and AL progression with RLRL than with LDA.[16,17] Notably, a previous crossover randomized controlled trial (RCT) reported marked and reversible choroidal thickening after RLRL, whereas 0.01% LDA showed minimal change (e.g., a mean increase of approximately 41 µm with RLRL versus 2 µm with LDA over 6 months).[17] These findings indicate potentially different treatment-associated responses between RLRL and LDA, which may be accompanied by distinct effects on posterior-pole morphology. However, the impact of these interventions on fine-scale 3D morphology of the retina and sclera remains largely unexplored.

Therefore, this study aimed to longitudinally compare changes in detailed macular curvature, MOSH, and derived shape asymmetry indices between children treated with RLRL and LDA from an RCT. These exploratory structural descriptors may provide insight into treatment-associated posterior-pole responses that are not captured by conventional axial-length and refractive outcomes alone.

Methods

Study Design and Participants

This study was a secondary analysis of a single-center, parallel-group RCT (ChiCTR2100045834). The methodology has been described previously.[16] Children aged 7 to 15 years with cycloplegic spherical equivalent refraction (SER) between −1.00 diopters (D) and −6.00 D, astigmatism ≤2.50 D, and best-corrected visual acuity (BCVA) ≥20/20 were enrolled. Exclusion criteria included previous myopia control treatment, anisometropia >1.50 D, or other ocular abnormalities. Fifty children with gradable OCT images at baseline and ≥1 follow-up visit were included in this morphological analysis.

Randomization and Interventions

Eligible children were randomly assigned to RLRL or LDA groups. The RLRL group used a desktop red-light device (650 ± 10 nm) for 3 minutes twice daily with ≥4-hour intervals. The LDA group instilled 0.01% atropine nightly. All wore single vision spectacles throughout the study. Follow-ups occurred at 1, 3, 6, and 12 months after baseline.

Ocular Examinations

Comprehensive examinations included autorefraction (ARK-1; Nidek) under cycloplegia at baseline, 6, and 12 months, and without cycloplegia at other visits. SER was calculated as the sphere plus half of the cylinder power. AL was measured using IOLMaster 500 (Carl Zeiss Meditec) before cycloplegia, and an average of at least 5 consistent measurements with an inter-measurement difference of ≤0.05 mm was recorded for analysis. Posterior segment imaging was performed using Cirrus HD-OCT 5000 (Carl Zeiss Meditec). The macular cube 512 × 128 scan protocol with enhanced depth imaging (EDI) mode acquired a 6 × 6 mm volumetric scan centered on the fovea for both eyes. For each eye-visit, morphological measurements were derived from a single gradable macular cube volume selected for segmentation. When more than one gradable cube was available for the same eye visit, the scan with the highest image quality, best foveal centration, and least motion artifact was used. Following image acquisition, the OCT device's integrated software automatically calculated standard retinal layer thickness parameters from the Early Treatment Diabetic Retinopathy Study (ETDRS) grid, including central foveal thickness, average ganglion cell-inner plexiform layer (GCL+IPL) thickness, average retinal nerve fiber layer (RNFL) thickness, and average outer retinal thickness. All layer segmentations were reviewed and manually corrected by a trained researcher.

Macular Curvature and Asymmetry

All OCT volumes were first assessed for quality by two independent graders masked to the participants' group allocation. Images were excluded if the signal strength index was <8, or if significant motion artifacts or poor foveal centration were present.

Figure 1 illustrates the complete workflow for measuring macular curvature and macular outward scleral height from OCT images. Gradable images were imported into custom-written software (OCTExplorer, version 3.8.0, Retinal Image Analysis Lab) for automated segmentation of the outer boundary of the retinal pigment epithelium/Bruch's membrane (RPE/BM) complex. The foveal center was defined as the point of minimum thickness between the inner limiting membrane and the RPE/BM. This OCT-based anatomic reference point was used for radial resampling and regional averaging. Because small localization differences may affect fovea-centered metrics in developing eyes, graders verified segmentation accuracy and foveal centration and performed manual correction when necessary; regional averages and symmetric meridian-based summaries were used to reduce, but not eliminate, the influence of small localization errors.

The segmented RPE/BM profile was exported to MATLAB (R2020b, MathWorks). Rather than averaging a fixed number of adjacent B-scans, curvature was derived from fovea-centered radial profiles resampled from the segmented macular cube. The RPE/BM coordinates along each of the 4 primary meridians (horizontal, vertical, 45 degrees, and 135 degrees) were resampled to generate 61 equidistant feature points at 0.1-mm intervals. To correct for optical distortions inherent in OCT imaging, a three-dimensional (3D) geometric transformation was applied using the eye's AL to estimate its radius of curvature, following the method described by Hwang et al.[18] A fifth-order polynomial was fitted to each meridian to provide an analytical approximation that balanced local contour flexibility with smooth first- and second-derivative estimation for curvature calculation, consistent with recent OCT-based radial curvature analyses.[19] Across all direction-level fits, the median R-squared was 0.999711 and the median root mean squared error (RMSE) was 0.002669. The curvature (K) at any point was

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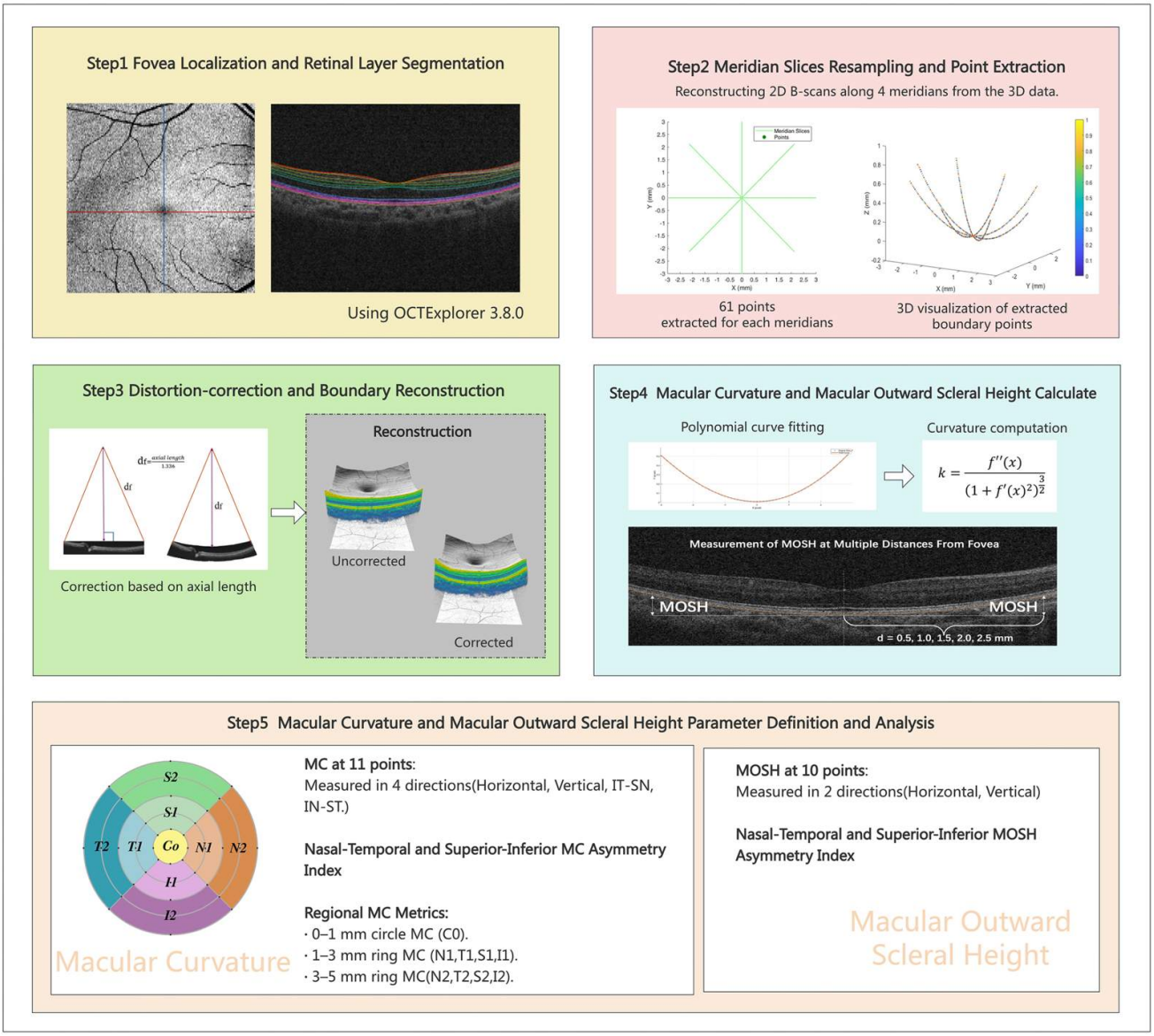


Figure 1. Flowchart for the measurement of macular curvature and macular outward scleral height from OCT images.

calculated using the formula:

$$K = \frac{f''(x)}{(1 + f'(x)^2)^{\frac{3}{2}}} \quad (1)$$

Curvature was subsequently calculated at 11 points (−2.5 to +2.5 mm from the fovea in 0.5-mm increments) along each meridian for further analysis. To ensure anatomic consistency, the 45 degres and 135 degrees meridians were standardized across left and right eyes to represent the inferonasal-to-superotemporal and inferotemporal-to-superonasal axes, respectively. Nasal-temporal (NT) and superior-inferior (SI) macular curvature asymmetry indices were calculated as the average difference in curvature between corresponding equidistant points.

For regional analysis, the macular area was divided into nine subfields using an ETDRS-like grid with custom diameters. These subfields were defined by the distance (r) from the foveal center as follows: a central zone ($r \leq 0.5$ mm), an inner ring (0.5 mm $< r \leq$ 1.5 mm), and an outer ring (1.5 mm $< r \leq$ 2.5 mm). The inner and outer rings were further subdivided into four quadrants (superior, nasal, inferior, and temporal). This yielded nine zones for analysis (see Fig. 1): a central zone (C0) and four inner (S1, N1, I1, and T1) and four outer (S2, N2, I2, and T2) quadrants. Additionally, a four-quadrant analysis was performed. Regional curvature was calculated as the average of all measurement points within each zone.

Macular Outward Scleral Height

We calculated MOSH to quantify posterior scleral deformation, following the method described by Li et al.[7,8] As shown in Figure 1, MOSH was defined as the vertical distance between a horizontal reference line passing through the subfoveal RPE/BM and the RPE/BM at predefined points of increasing eccentricity. We measured MOSH at 0.5, 1.0,

1.5, 2.0, and 2.5 mm from the fovea along the horizontal and vertical meridians. NT and SI asymmetry indices for MOSH were calculated similarly. MOSH was interpreted as an exploratory, height-based descriptor of posterior-pole deformation that complements curvature measures, rather than as a direct clinical outcome measure for myopia control.

Primary and Secondary Outcomes

The primary outcome was the 12-month change from baseline in average regional macular curvature. Secondary outcomes included changes in MOSH, the asymmetry indices for both curvature and MOSH, and OCT-derived retinal layer thicknesses.

Ethical Considerations

The study adhered to the Declaration of Helsinki and was approved by the Institutional Review Board of Peking University Shenzhen Hospital (No. 2021049A). Written informed consent was obtained from the parents.

Statistical Analysis

Longitudinal morphological outcomes were analyzed using linear mixed-effects models (LMMs) including both eyes if they met OCT quality criteria. Fixed effects included treatment group, visit, and the group-by-visit interaction, with baseline age, sex, and the baseline value of the corresponding outcome as covariates. Random intercepts for participants and for eyes nested within participants accounted for inter-individual variability, inter-eye correlation, and repeated measurements from the same eye. Missing data at specific visits were handled by the LMM procedure. Inter-group differences shown in the figures and tables were calculated as LDA minus RLRL. As age-sensitivity checks, associations of baseline age with baseline morphological metrics and longitudinal morphological changes were assessed using subject-clustered models and interpreted as sensitivity analyses rather than new endpoints. An additional age-restricted sensitivity analysis was performed among children aged 8 to 13 years at baseline to assess whether the main 12-month morphological findings were retained after narrowing the pediatric age range. Observed precision and detectable-effect estimates were derived from the model standard errors for the 12-month between-group contrasts. Conventional myopia-control outcomes were summarized for the morphological analysis cohort using clinical records from the parent trial; AL and SER changes were analyzed using Gaussian generalized estimating equations adjusted for baseline outcome, baseline age, and sex, with clustering by subject. Exploratory associations between AL change and morphological change were assessed at the eye level within treatment groups and interpreted descriptively. Statistical analyses were performed using Stata (version 18, StataCorp), with a 2-sided P value < 0.05 defined as statistically significant.

Results

Fifty children (25 per group) were included in the final morphological analysis, with a mean baseline age of 10.03 ± 1.71 years and equal sex distribution (50.0% girls). During follow-up, nine eye-visit OCT records were excluded from the corresponding morphological analyses after image-quality review. Baseline age was 9.97 ± 1.53 years in the RLRL group and 10.06 ± 1.93 years in the LDA group, and age was retained as a covariate in the longitudinal models. Supplementary Table S1 summarizes the baseline characteristics of the participants. The two groups were well-balanced at baseline, with no significant differences in age, sex, SER, AL, baseline regional macular curvatures, or central subfield retinal thickness (all $P > 0.05$).

Conventional Myopia-Control Outcomes in the Morphological Analysis Cohort

For the morphological analysis cohort, clinical records from the parent trial provided AL at baseline, 1, 3, 6, and 12 months, and SER at baseline, 6, and 12 months (Supplementary Table S2). At 12 months, AL increased by 0.045 mm in the RLRL group and 0.350 mm in the LDA group, with an adjusted LDA-minus-RLRL difference of 0.306 mm (95% confidence interval [CI] = 0.188 to 0.424, $P < 0.001$). SER change at 12 months was 0.065 D in the RLRL group and −0.630 D in the LDA group, with an adjusted difference of −0.709 D (95% CI = −0.975 to −0.443, $P < 0.001$). The direction of these conventional outcomes was consistent with less axial elongation and less SER progression in the RLRL group.

Longitudinal Point-Wise Changes in Macular Curvature

Over the 12 months, the RLRL group demonstrated significant and extensive relative flattening of the macular curvature compared to the LDA group. Supplementary Table S3 shows the longitudinal changes in point-wise macular curvature for both groups, after adjusting for baseline age, sex, and the baseline value of the corresponding outcome. At 12 months, this inter-group difference was statistically significant at 28 of the 40 (70%) measured points. This relative flattening effect was most pronounced near the fovea, observed at all points (100%) at 0.5-mm eccentricity, and decreased with distance: 87.5% of points at 1.0 mm, 75% at 1.5 mm, 62.5% at 2.0 mm, and 25% at 2.5 mm showed significant flattening in the RLRL group (Fig. 2; red triangles). The development of these inter-group differences was both time-dependent and location-specific. Initially, at 1 month, no significant differences in curvature change were observed at any point. By 3 months, significant relative flattening in the RLRL group began to emerge, primarily along the superior meridian. This effect expanded by 6 months, with significant points predominantly located in the superior, supero-nasal, and infero-nasal regions. At 12 months, the number of significant points peaked, but spatial distribution evolved. The superior meridian exemplifies this: initially a primary site of change, by 12 months, significant flattening remained only at 0.5 mm, whereas other meridians became more affected.

Changes in Regional Macular Curvature and Asymmetry Index

Regional analysis showed the RLRL group had significantly greater flattening than 0.01% LDA across all regions, particularly at later timepoints (Supplementary Fig. S1). No significant inter-group differences existed at 1 month but emerged across all 5 regions by 3 months and were maintained at 12 months. At 12 months, mean inter-group differences

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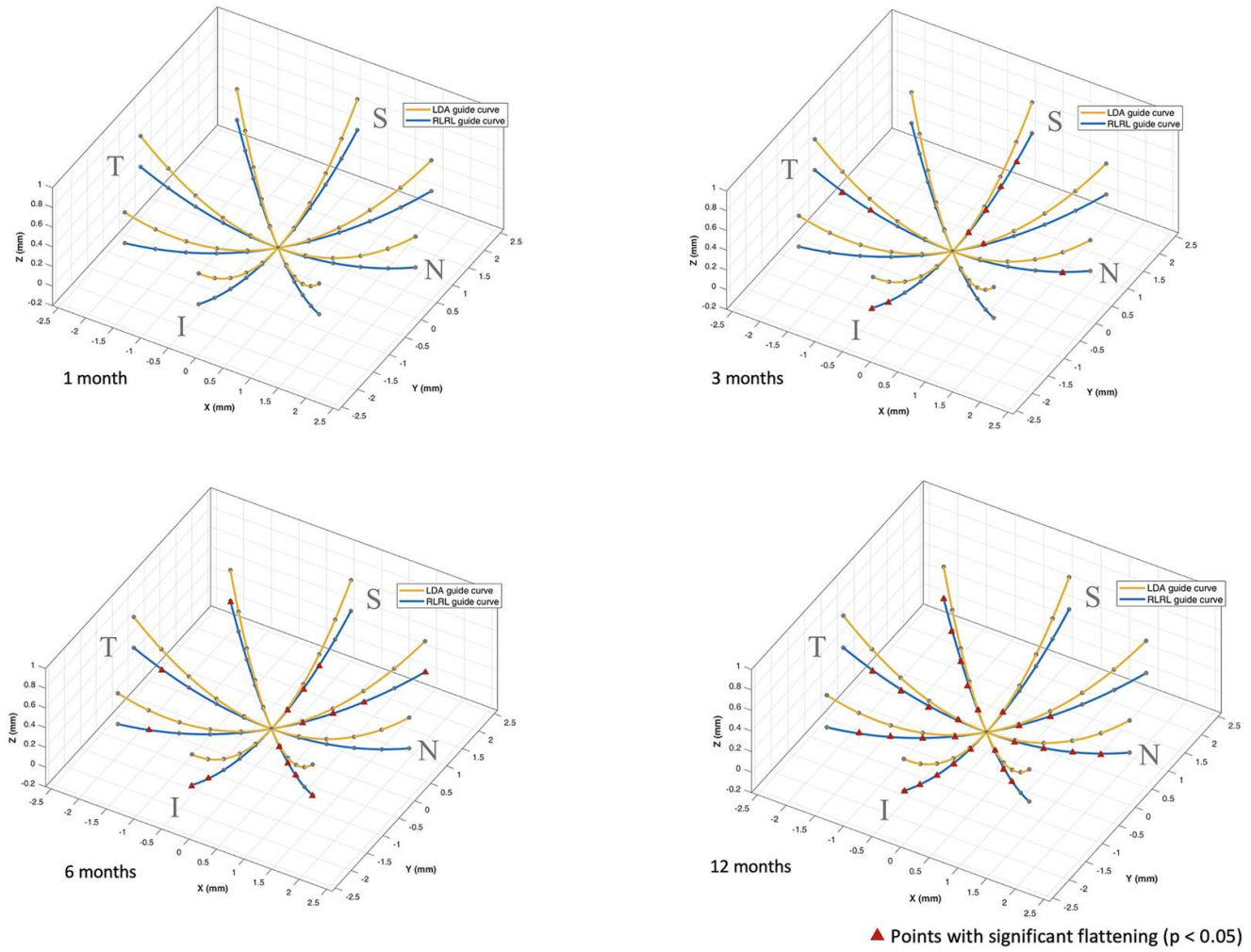


Figure 2. Comparison of point-wise macular curvature changes between the RLRL and LDA groups over 12 months, with points of significant relative flattening highlighted.

($\times 10^{-3}$ mm^{-1}), all favoring greater flattening in RLRL, were significant in the temporal quadrant (9.7, 95% CI = 6.7 to 12.8, $P < 0.001$), central zone (9.0, 95% CI = 4.4 to 13.6, $P < 0.01$), inferior quadrant (8.4, 95% CI = 5.2 to 11.6, $P < 0.001$), superior quadrant (7.0, 95% CI = 4.0 to 9.9, $P < 0.001$), and nasal quadrant (3.7, 95% CI = 0.8 to 6.7, $P < 0.05$).

In an age-restricted sensitivity analysis of children aged 8 to 13 years at baseline, 41 children were retained (22 RLRL and 19 LDA). The 12-month regional curvature showed the same direction as the full-cohort analysis; central, temporal, inferior, and superior regions remained nominally significant, whereas the smaller nasal difference was directionally consistent but less precise.

This consistent inter-group difference resulted from opposing longitudinal trends within each group (Fig. 3). The RLRL group showed progressive macular flattening from baseline, which became statistically significant in the central, temporal, and inferior regions by 12 months. Conversely, the LDA group exhibited macular steepening, significant in all 4 quadrants at the 12-month visit. The nasal quadrant was unique: whereas the RLRL group showed no significant flattening in this region, a significant inter-group difference still arose due to the pronounced steepening observed in the LDA group. The Table presents detailed results for the complete nine-subfield analysis.

Analysis of the asymmetry indices showed significant inter-group differences at 12 months for both NT and SI axes (see Supplementary Table S3): the RLRL group had a significantly greater increase in NT asymmetry (mean difference, $\times 10^{-3}$ mm^{-1}, −5.7, 95% CI = −8.7 to −2.6, $P < 0.001$) and SI asymmetry (−2.9, 95% CI = −5.8 to 0.0, $P < 0.05$).

Changes in Macular Outward Scleral Height and Asymmetry Index

The MOSH analysis showed significant location-specific differences in posterior scleral deformation between groups, particularly at 12 months (Supplementary Fig. S2). Most prominent inter-group differences occurred along the vertical meridian. In the superior retina, the LDA group showed significantly less reduction (or greater positive change) in MOSH compared to the RLRL group at all measured points from 0.5 mm to 2.5 mm. A similar, although less pronounced, pattern appeared in the inferior retina, where significant differences were observed at the 0.5 mm to 1.5 mm locations. No significant differences in

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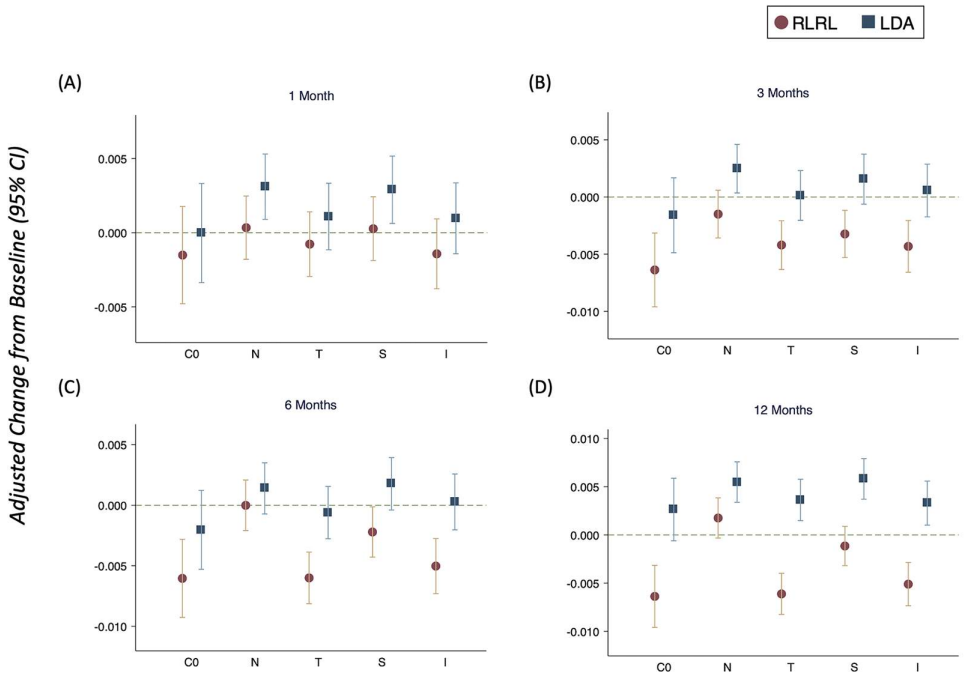


Figure 3. Adjusted changes in average regional macular curvature over 12 months in the RLRL and LDA groups: (**A**) 1 month; (**B**) 3 months; (**C**) 6 months; and (**D**) 12 months.

MOSH change occurred along the horizontal meridian at 12 months.

The within-group analysis at 12 months (Fig. 4) clarifies these findings. In the superior zone, the RLRL group exhibited a consistent decrease in MOSH from baseline (i.e. a reduction in scleral height). Conversely, in the inferior zone, the RLRL group showed a significant increase in MOSH from baseline at the 0.5 mm to 1.5 mm locations. The LDA group, however, showed minimal change from baseline across all meridians, with most values remaining near zero.

For MOSH asymmetry indices, significant differences emerged for NT at 3 months and for SI at 6 months, maintained at 12 months, with RLRL showing greater asymmetry changes (Supplementary Table S4).

Exploratory eye-level analyses showed that associations between AL change and regional curvature change were most apparent in the RLRL group, particularly for temporal curvature at 12 months (Pearson $r = 0.420$, $P = 0.003$) and inferior curvature at 12 months ($r = 0.502$, $P < 0.001$). The corresponding associations in the LDA group were weaker and not statistically significant, and the pattern was not uniform across all regions or MOSH asymmetry metrics. By contrast, exploratory analyses of visual function showed no robust association between 12-month corrected visual-acuity change and the main macular curvature or MOSH asymmetry metrics after correction for multiple exploratory comparisons and subject-level sensitivity analysis.

Changes in Retinal Layer Thicknesses

No significant differences existed between groups at any timepoint in adjusted changes for central foveal thickness, average GCL+IPL, average RNFL, or average outer retinal thickness (all $P > 0.05$; Supplementary Table S5).

Discussion

To our knowledge, this is the first longitudinal 3D morphological study to compare posterior-pole morphological profiles after RLRL and LDA therapy in myopic children. Our primary finding is that the two treatments were associated with distinct posterior-pole morphological profiles over 1 year. The RLRL group showed widespread macular flattening and non-uniform MOSH changes, whereas the LDA group showed comparatively stable posterior-pole morphology or slight steepening. Notably, the LDA group also showed substantially greater axial elongation and SER progression than the RLRL group in this cohort, and in the absence of an untreated control group, the slight steepening observed with LDA may at least partly reflect ongoing myopic progression rather than an atropine-specific morphological response. RLRL-associated flattening was also accompanied by a greater increase in macular curvature asymmetry compared with LDA at 12 months. No inter-group differences in retinal layer thickness changes were detected over this 12-month period.

RLRL-related vascular and choroidal responses may partly account for the posterior-pole morphological changes observed in this study. Previous studies have reported rapid choroidal thickening, often within the first month of treatment, and altered choroidal perfusion after RLRL therapy, findings that are compatible with treatment-associated changes in macular shape.[20–22] Beyond these vascular responses, experimental work on photobiomodulation suggests that red-light exposure can act through mitochondrial cytochrome c oxidase, increase cellular energy availability, reduce oxidative stress, and influence retinal dopamine signaling.[23–25] Müller glial responses, including Hippo-YAP pathway activation in experimental models, have also been proposed as a potential cellular link between photobiomodulation and tissue remodeling.[26] Although

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Table. Comparison of Adjusted Changes in Regional Macular Curvature Between Low-Level Red-Light and Low-Dose Atropine Groups Over 12 Months

Variables	RLRL Group Mean (95% CI) × 10^{-3} mm^{-1}	LDA Group Mean (95% CI) × 10^{-3} mm^{-1}	Difference Mean (95% CI) × 10^{-3} mm^{-1}
Average MC in area C0			
1 mo	−1.4 (−4.7 to 1.9)	0.1 (−3.3 to 3.4)	1.5 (−3.2 to 6.2)
3 mo	−6.3 (−9.6 to −3.1)	−1.6 (−4.8 to 1.7)	**4.8 (0.2 to 9.4)***
6 mo	−6.1 (−9.3 to −2.9)	−2.1 (−5.4 to 1.1)	4.0 (−0.6 to 8.6)
12 mo	−6.4 (−9.7 to −3.2)	2.6 (−0.7 to 5.8)	**9.0 (4.4 to 13.6)****
Average MC in area T1			
1 mo	−0.4 (−3.6 to 2.7)	−0.1 (−3.3 to 3.1)	0.4 (−4.1 to 4.9)
3 mo	−5.3 (−8.4 to −2.3)	−1.3 (−4.4 to 1.9)	4.1 (−0.3 to 8.5)
6 mo	−5.6 (−8.7 to −2.6)	−3.0 (−6.1 to 0.2)	2.7 (−1.7 to 7.1)
12 mo	−7.1 (−10.2 to −4.0)	1.8 (−1.3 to 4.9)	**8.9 (4.5 to 13.2)****
Average MC in area T2			
1 mo	−0.5 (−4.1 to 3.0)	3.1 (−0.6 to 6.8)	3.6 (−1.5 to 8.8)
3 mo	−1.5 (−5.0 to 1.9)	2.7 (−0.9 to 6.3)	4.2 (−0.8 to 9.2)
6 mo	−5.7 (−9.2 to −2.3)	3.5 (0.0 to 7.1)	**9.3 (4.3 to 14.3)****
12 mo	−3.5 (−7.0 to 0.0)	7.0 (3.5 to 10.6)	**10.5 (5.6 to 15.5)****
Average MC in area N1			
1 mo	−0.7 (−3.2 to 1.9)	2.1 (−0.5 to 4.7)	2.7 (−0.9 to 6.4)
3 mo	−2.6 (−5.1 to −0.2)	1.4 (−1.1 to 3.9)	**4.0 (0.5 to 7.6)***
6 mo	−2.2 (−4.7 to 0.2)	2.0 (−0.6 to 4.5)	**4.2 (0.7 to 7.7)***
12 mo	0.1 (−2.4 to 2.6)	6.5 (4.0 to 9.0)	**6.4 (2.9 to 9.9)****
Average MC in area N2			
1 mo	2.0 (−1.5 to 5.5)	5.2 (1.6 to 8.9)	3.2 (−1.8 to 8.3)
3 mo	1.8 (−1.6 to 5.3)	5.7 (2.2 to 9.2)	3.9 (−1.1 to 8.8)
6 mo	5.0 (1.6 to 8.5)	1.9 (−1.6 to 5.4)	−3.1 (−8.0 to 1.8)
12 mo	6.7 (3.2 to 10.1)	5.3 (1.8 to 8.8)	−1.4 (−6.2 to 3.5)
Average MC in area S1			
1 mo	0.4 (−2.2 to 3.0)	0.4 (−2.4 to 3.1)	−0.1 (−3.8 to 3.7)
3 mo	−3.8 (−6.3 to −1.4)	1.3 (−1.3 to 3.9)	**5.2 (1.6 to 8.8)***
6 mo	−3.4 (−5.9 to −0.9)	2.1 (−0.5 to 4.7)	**5.6 (2.0 to 9.2)***
12 mo	−1.1 (−3.5 to 1.4)	5.9 (3.4 to 8.5)	**7.0 (3.5 to 10.5)****
Average MC in area S2			
1 mo	0.1 (−3.4 to 3.7)	6.9 (3.0 to 10.7)	**6.8 (1.5 to 12.0)***
3 mo	−0.4 (−3.8 to 3.0)	2.9 (−0.7 to 6.6)	3.4 (−1.6 to 8.4)
6 mo	0.9 (−2.5 to 4.3)	1.7 (−1.9 to 5.3)	0.8 (−4.2 to 5.8)
12 mo	0.5 (−2.8 to 3.9)	5.8 (2.3 to 9.3)	**5.3 (0.4 to 10.1)***
Average MC in area I1			
1 mo	−2.0 (−5.1 to 1.0)	0.6 (−2.4 to 3.7)	2.7 (−1.6 to 6.9)
3 mo	−5.5 (−8.4 to −2.6)	−2.2 (−5.1 to 0.8)	3.3 (−0.8 to 7.5)
6 mo	−5.3 (−8.3 to −2.4)	−1.3 (−4.3 to 1.7)	4.0 (−0.2 to 8.2)
12 mo	−7.2 (−10.1 to −4.3)	2.9 (0.0 to 5.9)	**10.2 (6.0 to 14.3)****
Average MC in area I2			
1 mo	0.0 (−4.1 to 4.0)	1.9 (−2.2 to 6.1)	1.9 (−3.8 to 7.7)
3 mo	−1.4 (−5.3 to 2.5)	5.6 (1.6 to 9.6)	**7.0 (1.4 to 12.5)***
6 mo	−3.3 (−7.2 to 0.6)	3.3 (−0.7 to 7.3)	**6.6 (1.0 to 12.1)***
12 mo	−0.5 (−4.3 to 3.3)	6.0 (2.0 to 10.0)	**6.5 (1.0 to 12.0)***

CI, confidence interval; LDA, low-dose atropine; MC, macular curvature; RLRL, repeated low-level red light.
reported in mm^{-1} and scaled by 10^{-3} for change estimates.
Bold values indicate statistically significant between-group differences.
*$P < 0.05$; **$P < 0.001$.

these pathways were not measured here, they provide biological context for examining posterior-pole morphology alongside conventional refractive outcomes after RLRL.

Mechanically, RLRL-related choroidal expansion could alter local posterior-pole loading and, together with downstream retinal or scleral tissue responses, contribute to broader macular flattening.[27] This possibility is consistent with the idea that localized choroidal effects may be accompanied by wider macular shape change, and it may also help explain the non-uniform MOSH pattern observed here, including opposite directions of change in the superior and inferior meridians. It may further relate to previous reports of AL shortening after RLRL therapy, for which post hoc analyses have suggested that choroidal thickening alone accounts for only part of the measured AL reduction.[28–31] The present study did not directly test choroidal mediation, scleral biomechanics, or molecular pathways, and the morphological findings should not be taken as evidence for a specific choroidal, scleral, or cellular mechanism.

The interpretation of the LDA comparator also requires caution because this study used 0.01% atropine. Prior studies, including the LAMP study and subsequent clin-

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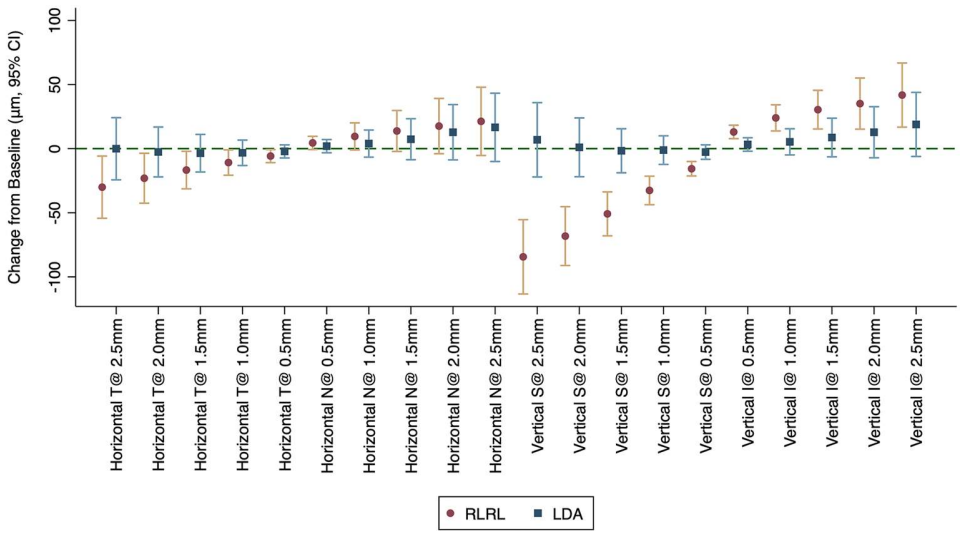


Figure 4. Adjusted changes in macular outward scleral height at discrete locations for the RLRL and LDA groups at 12 months.

ical and meta-analytic evidence, indicate that atropine-associated choroidal thickening is concentration-dependent, with smaller effects at 0.01% than at higher concentrations.[32–34] The morphological differences observed here should therefore not be generalized to all atropine concentrations. Thus, the between-group morphological differences observed here are inherently relative, reflecting the differential rather than absolute effects of each treatment. The independent contribution of each intervention and myopic progression to posterior-pole changes cannot be determined in this head-to-head design.

The temporal pattern further supports a gradual and spatially heterogeneous posterior-pole response. Macular curvature differences were absent at 1 month, emerged around 3 months, expanded by 6 months, and were most extensive by 12 months. In contrast, MOSH and asymmetry changes were more region-specific and less uniform over time. This sequence suggests that the observed morphological profile was not an immediate global shift of the posterior pole, but a progressive regional shape response that should be interpreted alongside conventional AL and SER outcomes.

The magnitude of the main curvature findings was also considered in relation to observed model precision. At 12 months, the observed differences for central, temporal, superior, and inferior curvature exceeded the approximate 80% detectable-difference threshold estimated from the model standard errors, whereas the nasal curvature difference was smaller than this threshold and is therefore interpreted more cautiously.

We observed greater increases in macular curvature asymmetry in the RLRL group at 12 months, despite overall flattening. Asymmetry is a recognized hallmark of myopic progression; studies have consistently shown that myopic eyes elongate non-uniformly, particularly along the nasal-temporal axis. The temporal choroid is often thinner than the nasal choroid, and its thinning correlates more strongly with axial elongation, identifying the temporal region as a key site of myopic stretching.[35,36] Furthermore, retinal curvature studies in highly myopic eyes have shown significant inter-quadrant differences, with the superior retina typically being the flattest and the nasal quadrant being flatter than the temporal and inferior quadrants.[11] In this context, the increased NT and SI macular curvature asymmetry indices observed with RLRL may reflect a spatially uneven treatment-associated shape response. The treatment effect was most pronounced in the central, temporal, and inferior regions, with minimal response in the nasal quadrant. However, without a non-treated control group, we cannot fully distinguish treatment-associated asymmetry change from natural progression of asymmetry in myopic children. Longer-term studies with larger sample sizes are needed to confirm these findings and determine the clinical implications of this increase in asymmetry.

Several limitations must be acknowledged. This was a secondary analysis with a 12-month follow-up, a modest sample size, no higher-dose atropine comparator, and no non-treated control group. This study did not include individual-level choroidal-thickness trajectories or visual-acuity trajectories for the morphological analysis cohort, and it did not distinguish choroidal from scleral contributions to the observed posterior-pole morphological changes. Pediatric foveal localization, including the possibility of shallow foveal pits in developing eyes, may also influence fovea-centered curvature and MOSH metrics despite segmentation and centration review and regional averaging. Although the age-restricted sensitivity analysis supported the direction of the main regional curvature findings, residual age-related differences in foveal anatomy and localization cannot be fully excluded. In addition, our analysis was constrained by the technical limits of spectral-domain OCT and meridional two-dimensional curvature measurements. These limitations underscore the need for future studies with longer-term follow-up to determine the clinical significance of these shape changes, particularly the increased asymmetry with RLRL. Validating these findings in larger, more diverse cohorts, ideally using a multi-modal approach that combines wide-field imaging with functional assessments, will be a critical next step.

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Conclusions

RLRL and 0.01% atropine were associated with distinct posterior-pole morphological profiles over 12 months, with RLRL showing broader macular flattening and the LDA group showing slight steepening. As this trial compared two active treatments without an untreated control, further studies are needed to clarify the different treatment effects on posterior-pole changes and ongoing myopic progression, as well as the underlying choroidal and scleral mechanisms.

Acknowledgments

The authors thank the InnoHK HKSAR Government for providing valuable support. The research work described in this paper was conducted in the JC STEM Lab of Innovative Light Therapy for Eye Diseases, funded by The Hong Kong Jockey Club Charities Trust.

Supported by the Global STEM Professorship Scheme (P0046113), Research Matching Grant Scheme (P0048181), and Start-up Fund for RAPs under the Strategic Hiring Scheme (P0048638) from HKSAR, Henry G. Leong Endowed Professorship in Elderly Vision Health, and Nanshan District Health Science and Technology Program (Shen Nan Ke [2021] NS080).

Author Contributions: YC and MH conceived and designed the study. ML, JL, and YL contributed to data acquisition and management. ZZ and YW graded the quality of the OCT scans, designed the custom MATLAB algorithm, and reviewed and corrected the segmentation results. ZZ performed the literature search and analyzed the data. ZZ, FS, RC, and YW contributed to the interpretation of the data. ZZ wrote the original draft of the manuscript. All authors critically revised the manuscript for important intellectual content and approved the final version.

Data Sharing Statements: The data used for this study are not openly available due to reasons of privacy and are available from the corresponding author upon reasonable request.

Disclosure: **Z. Zhao**, None; **Y. Wu**, None; **F. Song**, None; **M. Li**, None; **Y. Wang**, None; **R. Chen**, None; **Y. Liu**, None; **J. Li**, None; **M. He**, US11420072 (P), CN201910490186.6 (P), AU2020233703C1 (P), Eyerising International Pty Ltd. (E, I); **Y. Chen**, None

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